

1.1 Tools and Resources Required

➤ Database: MySQL

XAMPP is an open-source local development environment created by Apache Friends. It includes Apache HTTP Server, MySQL Database, and interpreters for PHP and Perl, making it ideal for web development. MySQL will be used to manage all user data, skill records, and barter requests securely. XAMPP allows for easy setup and local testing before deployment (DSouza, 2024).



Figure 1: Xampp

➤ Framework: Laravel

The Model-View-Controller (MVC) design is used by the open-source PHP framework Laravel. It ensures organized and effective code by offering built-in features for database management, routing, and authentication. Laravel will be utilized in this project to handle data movement between the front end and back end, provide the fundamental features of the Skill Barter System, and uphold system security (tutorialspoint, 2024).



Figure 2: Laravel

➤ **Frontend: HTML, CSS, and JavaScript**

JavaScript adds logic and interactivity to improve user experience, CSS improves layout and design, and HTML establishes the webpages' structure. Together, they guarantee that the Skill Barter System is responsive, aesthetically pleasing, and simple to use on desktop and mobile platforms.



figure 3: HTML,CSS & JS

➤ **Backend: PHP**

PHP is a strong programming language that is used to manage backend operations including database communication, barter request processing, and user authentication. It ensures smooth operation between client and server, making the platform secure and dynamic (WillDom, 2023).



figure 3: PHP

➤ **Text Editor: Visual Studio Code**

Visual Studio Code, a free and open-source code editor, supports multiple programming languages. It facilitates the development of the Skill Barter System by providing features including real-time collaboration, debugging tools, and syntax highlighting (UCHENDU, 2021).



figure 4: VS Code

➤ **Documentation: Microsoft Word**

The Skill Barter System will use Microsoft Word to create reports, documentation, and proposals. It ensures that written documents are presented professionally by enabling structured formatting, editing, and integration of figures and tables.



figure 5: MS word 365

➤ **Design Tools: Figma and Draw.io**

System flowcharts, ER diagrams, and UML models will be produced using Draw.io (also called Diagrams.net). Diagramming project designs is made easier by its support for cloud storage and data privacy (Edirisinghe, 2023).

Figma will be used for UI/UX design. It is a collaborative online tool that allows for wireframing and interface design, helping create a modern and user-friendly look for the Skill Barter System (Purdila, 2023).



figure 6: Figma



figure 7: Draw.io